F1 — Baseline Foaming Cleanser — 1 L Batch Card

Goal: Simple, clear, stable; good foam; VOC-free; BOV-ready. | Target pH: 5.8–6.0 | Viscosity: ≈250–400 cP

Prep & Safety

- ☐ PPE (goggles, nitrile gloves, apron)
- ☐ Calibrate/verify pH meter; prep 10% citric solution
- ☐ Warm bath ready (70°C) for SCI dissolve / surfactant phase

Ingredients (grams for 1 L)

- Water (hold back ~50 g): q.s. (~980 g)
- SCI: 40 g
- CAPB: 60 g
- Capryl Glucoside: 20 g
- Polysorbate 20: 10 g
- Vegetable Glycerin: 20 g
- Sodium Phytate: 2 g
- Peppermint EO: 3 g
- Preservative: 10 g (Optiphen) / 8 g (Phenoxy)
- Citric 10% soln: to pH

Process

1. Phase A: Water + glycerin at 70°C; slurry SCI and dissolve with low shear.
2. Add CAPB, then Capryl Glucoside; mix gently to avoid aeration.
3. Cool to <40°C. Add Polysorbate 20, chelator, then fragrance.
4. Adjust pH to 5.8–6.0 with 10% citric acid solution.
5. Add preservative; mix 5 min.
6. De-gas (stand or gentle vacuum). Filter 100–200 µm before BOV fill.

QC & Records

- ☐ Measure pH (target 5.8–6.0)
- ☐ Measure viscosity (Brookfield LV, spindle 2/3 @20 rpm)
- ☐ Foam test: cylinder 10 inversions (record 0/1/5 min)
- ☐ Grease-cut test: spray, 30s dwell, wipe rating
- ☐ De-gassed & filtered (100–200 µm)
- ☐ BOV fill sample labeled (code, date, batch#)

Batch Code: ____________________ Date: ___________ Operator: ___________

Notes: ____

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F2 — Boosted Cleaning (Hydrotrope + Co■solvent) — 1 L Batch Card

Goal: Better grease/soil tolerance; remains BOV-sprayable. | Target pH: 5.8–6.0 | Viscosity: ≈250–450 cP

Prep & Safety

- ☐ PPE (goggles, nitrile gloves, apron)
- ☐ Calibrate/verify pH meter; prep 10% citric solution
- ☐ Warm bath ready (70°C) for SCI dissolve / surfactant phase

Ingredients (grams for 1 L)

- Water q.s.: to 1000 g
- SCI: 35 g
- CAPB: 60 g
- Capryl Glucoside: 20 g
- SXS (40%): 20 g
- DPG: 20 g
- Sodium Gluconate: 3 g
- Polysorbate 20: 10 g
- Vegetable Glycerin: 20 g
- Peppermint EO: 3 g
- Preservative: 10 g (Optiphen) / 8 g (Phenoxy)
- Citric 10% soln: to pH

Process

1. Phase A: Water + glycerin at 70°C; slurry SCI and dissolve with low shear.
2. Add CAPB, then Capryl Glucoside; mix gently to avoid aeration.
3. Cool to <40°C. Add Polysorbate 20, chelator, then fragrance.
4. Add SXS (hydrotrope) and any co■solvents (e.g., DPG/Propylene Carbonate).
5. Adjust pH to 5.8–6.0 with 10% citric acid solution.
6. Add preservative; mix 5 min.
7. De-gas (stand or gentle vacuum). Filter 100–200 µm before BOV fill.

QC & Records

- ☐ Measure pH (target 5.8–6.0)
- ☐ Measure viscosity (Brookfield LV, spindle 2/3 @20 rpm)
- ☐ Foam test: cylinder 10 inversions (record 0/1/5 min)
- ☐ Grease-cut test: spray, 30s dwell, wipe rating
- ☐ De-gassed & filtered (100–200 µm)
- ☐ BOV fill sample labeled (code, date, batch#)

Batch Code: ____________________ Date: ___________ Operator: ___________

Notes: ____

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F3 — Heavy■Duty Anionic System — 1 L Batch Card

Goal: Stronger cleaning; clearer rinse; anionic■forward. | Target pH: 5.8–6.0 | Viscosity: ≈300–600 cP

Prep & Safety

- ☐ PPE (goggles, nitrile gloves, apron)
- ☐ Calibrate/verify pH meter; prep 10% citric solution
- ☐ Warm bath ready (70°C) for SCI dissolve / surfactant phase

Ingredients (grams for 1 L)

- Water q.s.: to 1000 g
- AOS■40: 80 g
- CAPB: 40 g
- SCI: 20 g
- Capryl Glucoside: 20 g
- SXS (40%): 20 g
- HEC: 2 g
- Sodium Citrate: 2 g
- Chelator: 2–3 g
- Polysorbate 20: 8 g
- Peppermint EO: 2.5–3 g
- Preservative: 10 g (Optiphen) / 8 g (Phenoxy)
- Citric 10% soln: to pH

Process

1. Disperse HEC at room temp; allow full hydration 15–30 min before heating.
2. Phase A: Water + glycerin at 70°C; slurry SCI and dissolve with low shear.
3. Add CAPB, then Capryl Glucoside; mix gently to avoid aeration.
4. Cool to <40°C. Add Polysorbate 20, chelator, then fragrance.
5. Add SXS (hydrotrope) and any co■solvents (e.g., DPG/Propylene Carbonate).
6. Adjust pH to 5.8–6.0 with 10% citric acid solution.
7. Add preservative; mix 5 min.
8. De-gas (stand or gentle vacuum). Filter 100–200 µm before BOV fill.

QC & Records

- ☐ Measure pH (target 5.8–6.0)
- ☐ Measure viscosity (Brookfield LV, spindle 2/3 @20 rpm)
- ☐ Foam test: cylinder 10 inversions (record 0/1/5 min)
- ☐ Grease-cut test: spray, 30s dwell, wipe rating
- ☐ De-gassed & filtered (100–200 µm)
- ☐ BOV fill sample labeled (code, date, batch#)

Batch Code: ____________________ Date: ___________ Operator: ___________

Notes: ____

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F4 — Advanced (Builders + Polymer Anti■redeposition) — 1 L Batch Card

Goal: Highest performance; watch clarity/viscosity & filtration. | Target pH: 5.8–6.0 | Viscosity: ≈300–600 cP

Prep & Safety

☐ PPE (goggles, nitrile gloves, apron)
☐ Calibrate/verify pH meter; prep 10% citric solution
☐ Warm bath ready (70°C) for SCI dissolve / surfactant phase

Ingredients (grams for 1 L)

- Water q.s.: to 1000 g
- AOS■40: 70 g
- CAPB: 50 g
- Capryl Glucoside: 20 g
- SCI: 20 g
- SXS (40%): 20 g
- Propylene Carbonate: 10 g
- DPG: 15 g
- Sodium Polyacrylate: 2 g
- Chelator: 2–3 g
- Polysorbate 20: 8 g
- Peppermint EO: 2.5–3 g
- Preservative: 10 g (Optiphen) / 8 g (Phenoxy)
- Citric 10% soln: to pH

Process

1. Phase A: Water + glycerin at 70°C; slurry SCI and dissolve with low shear.
2. Add CAPB, then Capryl Glucoside; mix gently to avoid aeration.
3. Cool to <40°C. Add Polysorbate 20, chelator, then fragrance.
4. Add SXS (hydrotrope) and any co■solvents (e.g., DPG/Propylene Carbonate).
5. Add Sodium Polyacrylate below 35°C; allow full hydration before pH/preservative.
6. Adjust pH to 5.8–6.0 with 10% citric acid solution.
7. Add preservative; mix 5 min.
8. De-gas (stand or gentle vacuum). Filter 100–200 µm before BOV fill.

QC & Records

☐ Measure pH (target 5.8–6.0)
☐ Measure viscosity (Brookfield LV, spindle 2/3 @20 rpm)
☐ Foam test: cylinder 10 inversions (record 0/1/5 min)
☐ Grease-cut test: spray, 30s dwell, wipe rating
☐ De-gassed & filtered (100–200 µm)
☐ BOV fill sample labeled (code, date, batch#)

Batch Code: ____________________ Date: ___________ Operator: ___________

Notes: ____

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